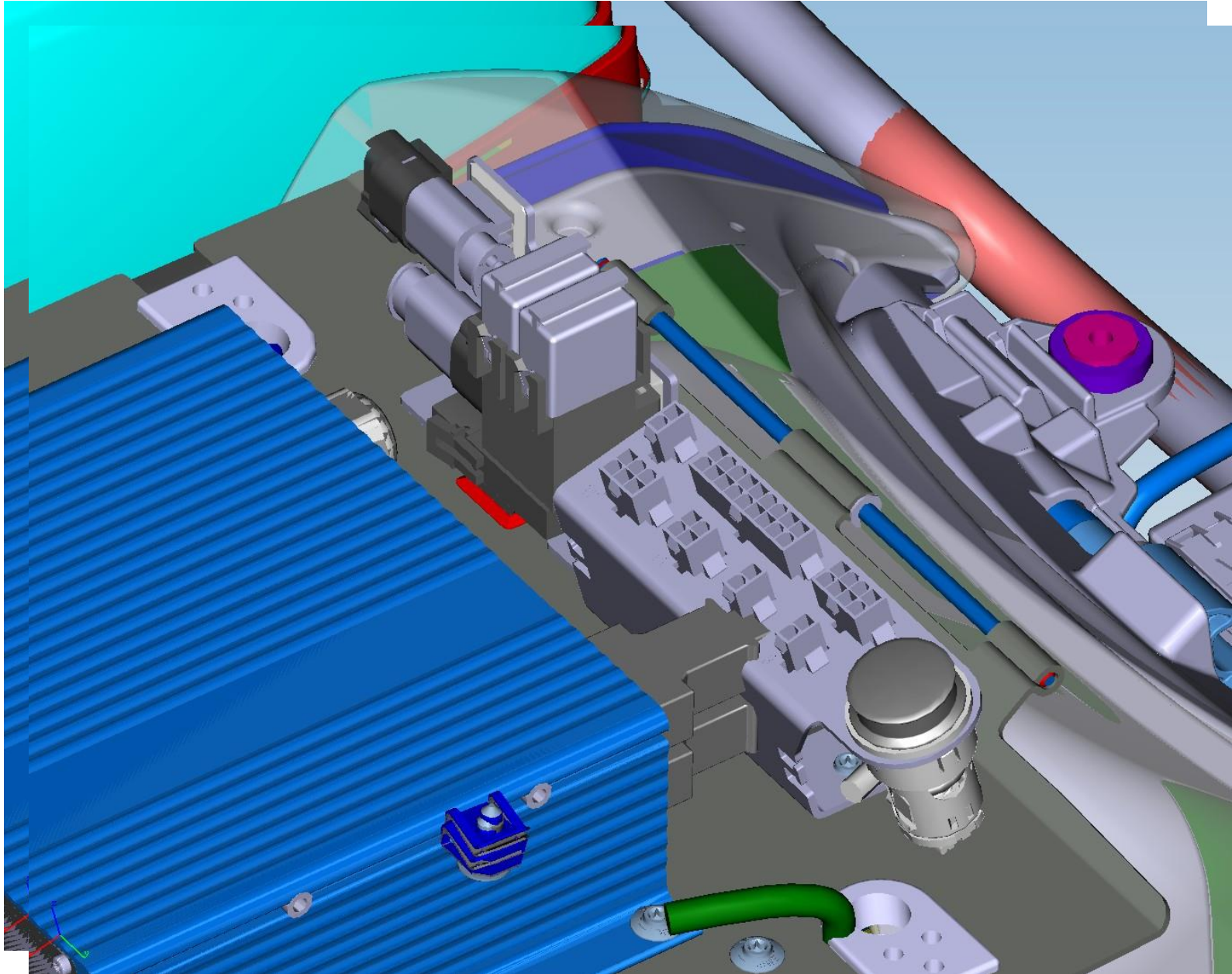


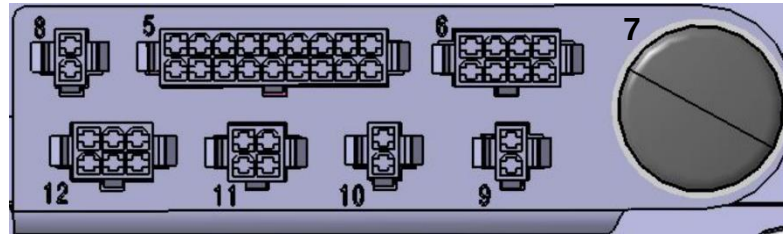
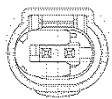
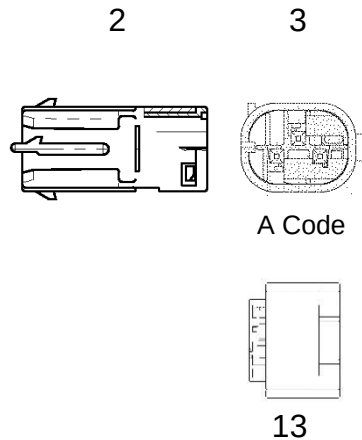
# R 1200 / 1250 RT-P Harness Connections

MY 2015 forward

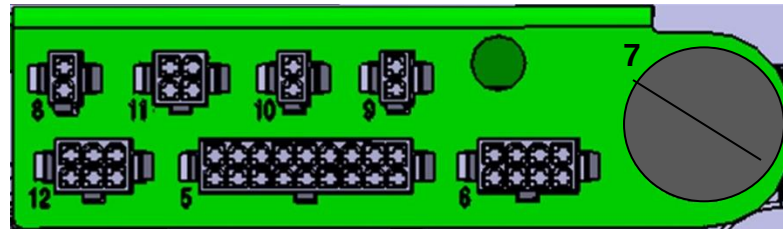


# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward



MY15-18



MY19+

- 1-Gun Lock (6A) (under radio box floor plate)
- 2-Radio (15A)
- 3-Auxiliary Rear LED Lights (mounting location only)
- 4-Relays (**radio power**, **radio speaker cut-out**)
- 5-LED Flasher Connection (10A + 6A)
- 6-Helmet Headset Interface (2A)
- 7-Lighter Socket (6A)
- 8-Accessory I (2A)
- 9-Accessory II (2A)
- 10-Accessory III (2A)
- 11-Radar Connection (6A)
- 12-Fairing Speakers
- 13-Euro PA Microphone Connection (parallel connection to PA in plug #6)

# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward

### Radio Box Connections

#### 1-Gun Lock, 6A

X1716 <https://fmsaccessories.com/sealed-bmw-connector-sets>



- 1 Power Gun Lock (from Pin 46 – 6A, 30 seconds after activation) GR RS
- 2 Ground BR

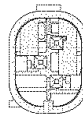
#### 2-Radio, 15A

ET865 <https://fmsaccessories.com/radio-connection-plug-code-a>



- 1 Radio Power + (from relay controlled by GMSF Pin 9, 0.3A, 30 min. after ign. off) RT SW
- 2 Radio Ground BR

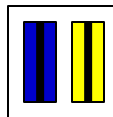
#### 3-Auxiliary Rear Duplex LED (holder clip only)



See Flasher Diagram

#### 4-Relays

K73



Relays for **Radio** & **Fairing Speaker Cut-out**

#### 8-Accessory I

ET884 <https://fmsaccessories.com/accessory-i-iii-connection-plugs>



- 1 Auxiliary I (GMSF Pin 33 – 2A, 60 seconds after ign. off) GN RT
- 2 Ground BR

#### 5-LED Flasher Input

ET381



See Flasher Diagram

#### 6-Helmet Headset Interface

ET1715 <https://fmsaccessories.com/helmet-headset-connection-plugs>



- 1 F1 PTT Switch Signal (GMSF Pin 23 – 0.65A, momentary, 30 minutes after ign. off) SW GR
- 2 F2 PTPA Switch Signal (GMSF Pin 34 – 2A, momentary, 30 minutes after ign. off) SW RT
- 3 Headset Interface Power (GMSF Pin 20 – 2A, 30 minutes after ign. off) SW VI
- 4 Ground - BR
- 5 Side Stand Extended Signal (direct connection to MC side stand switch in harness ) GR GN
- 6 Microphone NF to Siren PA
- 7 PTT + to siren PA (do not activate directly from F2 pin 2 or increased noise will result)
- 8 PTT 2 Switch Signal (GMSF Pin 21 – 0.6A, momentary; 30 minutes after ign. off) SW GN

# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward

### Radio Box Connections

#### 7-Accessory Lighter Socket

ET1427 <https://fmsaccessories.com/printr-power-connector>



- 2 Accessory Power (GMSF Pin 3 – 6A, 30 minutes after ign. off / shared with Radar Plug) RT VI
- 1 Ground BR

#### 9-Accessory II

ET1711 <https://fmsaccessories.com/accessory-i-iii-connection-plugs>



- 1 Auxiliary II (GMSF Pin 35 – 2A, 25 minutes after ign. off) GN WS
- 2 Ground BR

#### 10-Accessory III

ET885 <https://fmsaccessories.com/accessory-i-iii-connection-plugs>



- 1 Auxiliary III (GMSF Pin 36 – 2A, 30 minutes after ign. off) GN BL
- 2 Ground BR

#### 11-Radar, 6A

ET1712 <https://fmsaccessories.com/radar-connection-plug>



- 1 Radar Power (from Pin 3 – 6A, 30 minutes after ign. off) RT VI
- 2 Ground - BR
- 3 Vehicle Speed Signal GR GE (harness line contains a 10KΩ resistor)
- 4 Siren Video Trigger Signal + GR RT

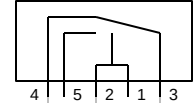
#### 12-Fairing Speakers

ET380 <https://fmsaccessories.com/speaker-connection-plugs>



- 1 LH Fairing Speaker + GE RT
- 2 LH Fairing Speaker - GE BR
- 4 RH Fairing Speaker + BL GE
- Fairing Speaker Cut-Out BL VI (GMSF Pin 7 – 2A, 30 minutes after ign. off)
- 5 RH Fairing Speaker - BL BR

Fairing Speaker Relay  
Normally Closed



- 3 Auxiliary Wires BR GE
- 6 Auxiliary Wires SW GE



X1717  
Bundled by front  
right frame by horn  
mount.

TE  
2-968790-1

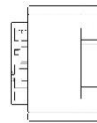
# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward

### Other Electrical Connections

#### 13-Euro PA Microphone Connection Plug

<https://fmsaccessories.com/sealed-bmw-connector-sets>



- 1 Supply Voltage Power + GR SW
- 2 Push-To-Talk BL GE
- 3 Microphone NF to Siren PA SW VI
- 4 Ground - WS BR

#### Front Accessory USB Socket (dealer installed option)

<https://fmsaccessories.com/fairing-usb-outlet-w-voltmeter-on-off-switch>

Cigarette  
Lighter  
or USB  
Socket



- 1 Ground - BR
- 2 Front Power Plug + (duplicate from front power socket or fused harness) RT GN

#### Front Lighter / USB Socket

X98 <https://fmsaccessories.com/fairing-waterproof-lighter-socket>



- 1 Ground - BR
- 2 Front Power Plug + (duplicate from front power socket) RT GN

#### Fused Front Socket Harness

(dealer installed option)

<https://fmsaccessories.com/front-fused-power-socket-harness>



- 1 Ground - BR
- 2 Front Power Plug + (Fused Direct Power from Main Battery) RT

#### Map Light Plug @ Steering Head

ET1718 <https://fmsaccessories.com/sealed-bmw-connector-sets>



- 1 Map Light + (GMSF Pin 47 - 2A, 30 minutes after ign. off) GE
- 2 Ground - BR

#### Mating BMW Power Socket Plug

<https://fmsaccessories.com/power-socket-male-plug>



Connect to heated clothing or other 12v DC electrical needs

# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward

### LED Lighting System

#### Light Connections Inside Rear Housing on Flasher Board

- LH Rear Side LED
- LH Rear LED / Cruise
- RH Rear LED / Cruise
- RH Rear Side LED
- LH Side Turn Signal
- RH Side Turn Signal
- Rear Brake / Tail Light
- Rear Blue ID LED Lights
- Saddlebag Lights

#### Flasher Dip Switch Map

- 1-Flash Actuation Grouping
- 2-Enable Alley Lights with warning Lights
- 3-CA LH Front Warning Light
- 4-Bootloader (to reprogram if needed)



Rear Lights

LED  
Flasher in  
Rear  
Housing



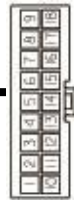
#### Video Connection Plug

<https://fmsaccessories.com/video-system-icon-plug>

- 1-LH Turn Signal
- 2-RH Turn Signal
- 3-Rear Brake
- 4-Rear Emergency Lights

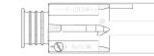
2

6



ET381

- 1 LH Auxiliary Rear LED Light
- 2 RH Auxiliary Rear LED Light
- 3 Ground



Rear Auxiliary Light



- 1 LH Front Side LED SW BL 1
- 2 LH Alley LED SW BR 2
- 3 LH Front Forward LED / Cruise SW GE 3
- 4 LH Front Take Down Light SW GN 4
- LH Ground BR 7
- Not Used 5, 6, 8

- 5 RH Front Side LED WS BL 1
- 6 RH Alley LED WS BR 2
- 7 RH Front Forward LED / Cruise WS GE 3
- 8 RH Front Take Down Light WS GN 4
- 9 Rearward LED Indicator Light (red) GE OR 5
- 10 Forward LED Indicator Light (amber) GE VI 6
- RH Ground BR 7
- Not Used 8

11 LIN-bus  
VI WS

- InterClear® Signal from Siren
- Cruise Light Switch (I)
- Alley Light LH Switch (C)
- Alley Light RH Switch (D)
- LED Light Activation (P)
- Take Down Light – Steady Burn (E)
- Take Down Light – Wig Wag (F)

- 12 Emergency Light Power Source (GMSF Pin 5, 10A), 30 minutes after ign. off GN RT
- 13 Take Down Light Power Source (GMSF Pin 2, 6A), 30 minutes after ign. off GR RT
- 14 Brake / Tail Light GR SW GE
- 15 Left Side Turn Signal BL RT
- 16 Right Side Turn Signal BL SW
- 17 Ground BR
- 18 Ground BR

Left Front Lights



TE  
106527-8

Right Front Lights



TE  
106527-8

# R 1200 / 1250 RT-P Harness Connections

## MY 2015 forward

### LED Lighting System – LAPD Harness

#### Light Connections Inside Rear Housing on Flasher Board

- LH Rear Side LED
- LH Rear LED / Cruise
- RH Rear LED / Cruise
- RH Rear Side LED
- LH Side Turn Signal
- RH Side Turn Signal
- Rear Brake / Tail Light
- Rear Blue ID LED Lights
- Saddlebag Lights

#### Flasher Dip Switch Map

- 1-Flash Actuation Grouping
- 2-Enable Alley Lights with warning Lights
- 3-CA LH Front Warning Light
- 4-Bootloader (to reprogram if needed)



Rear Lights

**LED Flasher in Rear Housing**

ET381

Connect with LH Rear Side LED

Connect with RH Rear Side LED

2

6

- 1 LH Auxiliary Rear LED Light
- 2 RH Auxiliary Rear LED Light
- 3 Ground



|    |                                                                              |    |
|----|------------------------------------------------------------------------------|----|
| 1  | LH Front Side LED + RH Rear Side LED BL SW                                   | 5  |
| 2  | LH Alley LED SW RT                                                           | 2  |
| 7  | LH Front Forward LED / Cruise SW WS                                          | 3  |
| 8  | LH Front Take Down Light SW                                                  | 4  |
|    | LH Ground BR                                                                 |    |
|    | Not Used                                                                     |    |
| 5  | RH Front Side LED + LH Rear Side LED WS SW                                   | 1  |
| 6  | RH Alley LED RT                                                              | 6  |
| 3  | RH Front Forward LED / Cruise WS                                             | 7  |
| 4  | RH Front Take Down Light GN                                                  | 8  |
| 9  | Rearward LED Indicator Light (red) OR                                        | 9  |
| 10 | Forward LED Indicator Light (amber) OR SW                                    | 10 |
|    | RH Ground BR                                                                 |    |
|    | Not Used                                                                     |    |
| 11 | LIN-bus RT WS                                                                | 11 |
|    | InterClear® Signal from Siren                                                |    |
|    | Cruise Light Switch (I)                                                      |    |
|    | Alley Light LH Switch (C)                                                    |    |
|    | Alley Light RH Switch (D)                                                    |    |
|    | LED Light Activation (P)                                                     |    |
|    | Take Down Light – Steady Burn (E)                                            |    |
|    | Take Down Light – Wig Wag (F)                                                |    |
| 12 | Emergency Light Power Source (GMSF Pin 5, 10A), 30 minutes after ign. off GR | 12 |
| 13 | Take Down Light Power Source (GMSF Pin 2, 6A), 30 minutes after ign. off VI  | 13 |
| 14 | Brake / Tail Light BL                                                        | 14 |
| 15 | Left Side Turn Signal BL WS                                                  | 15 |
| 16 | Right Side Turn Signal RT SW                                                 | 16 |
| 17 | Ground BR                                                                    | 17 |
| 18 | Ground BR WS                                                                 | 18 |

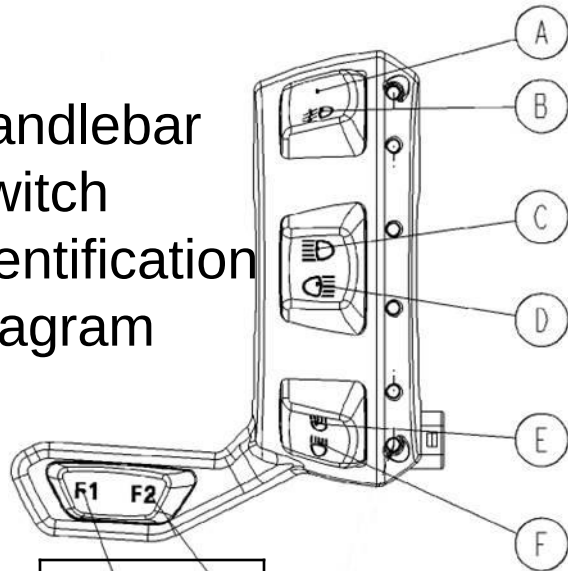


# General Module SF Controller System Output Allocation

| Pin Location | Amps  | Running Time* | Description                                                                                    |
|--------------|-------|---------------|------------------------------------------------------------------------------------------------|
| 34           | 2A    | 30 min        | F2, PTPA Switch Output (H) SW/RT x1715                                                         |
| 35           | 2A    | 25 min        | Accessory II Power Source GN WS x1711                                                          |
| 36           | 2A    | 30 min        | Accessory III Power Source GN BL X885                                                          |
| 47           | 2A    | 30 min        | Map Light GE x1718                                                                             |
| 2            | 6A    | 30 min        | TDL Power Source GR RT A381                                                                    |
| 1            | 6A    | 30 min        | Siren Power (power stays on 30 min for PA)                                                     |
| 9            | 0.3A  | 30 min        | Radio Power Relay (FUNK) BL WS x865                                                            |
| 8            | 0.6A  | 30 min        | Power Supply P-Switches                                                                        |
| 21           | 0.6A  | 30 min        | <a href="#">PTT 2 (license reading)</a> SW GN x1715                                            |
| 3            | 6A    | 30 min        | Radar Power Source + Cig. Lighter RT VI x1712                                                  |
| 22           | 0.3A  |               | <a href="#">Not Used</a>                                                                       |
| 23           | 0.65A | 30 min        | F1, PTT Output (G) SW GR x1715                                                                 |
| 7            | 2A    | 30 min        | Radio Speaker Mute Switch output to Speaker Relay                                              |
| 20           | 2A    | 30 min        | Helmet Headset Interface Power Source SW VI x1715                                              |
| 46           | 6A    | 30 sec        | Gun Lock Power Source (hold power on for 30 seconds – lapse after ignition is off) GR RS x1716 |
| 37           |       |               | Authority Relay                                                                                |
| 51           | 0.3A  |               | Battery Voltage Measurement                                                                    |
| 5            | 10A   | 30 min        | Emergency Lights Power Source GN RT a381                                                       |
| 33           | 2A    | 30 sec        | Accessory I Power Source + Radio Box Power Unlocking System GN RT x884                         |
|              |       |               |                                                                                                |

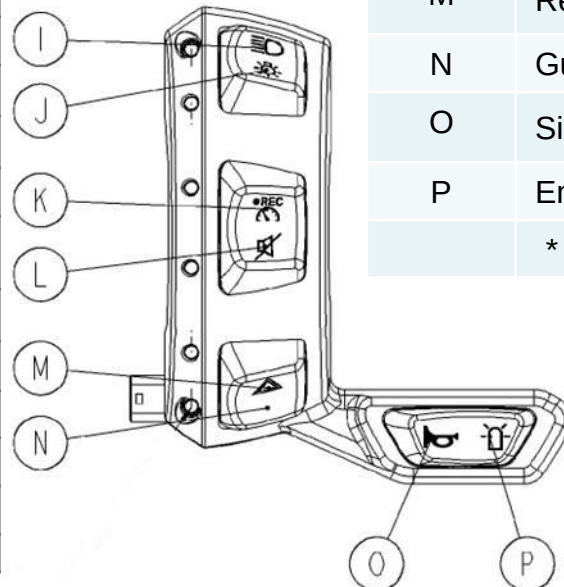
\*after ignition off

# Handlebar Switch Identification Diagram



BU Behörde USA  
(c) BC1,BC2,BC3,BC5, BC9,  
BC20, BC35, BC36, BC47

|                      |                |
|----------------------|----------------|
|                      | X113A          |
| A (Taster)           | (T) BC21       |
| B (Taster)           | wie Serie      |
| C (Schalter rastend) | auf LIN        |
| D (Schalter rastend) | auf LIN        |
| E (Schalter rastend) | auf LIN        |
| F (Schalter rastend) | auf LIN        |
| (F1) G (Taster)      | (T) BC23       |
| (F2) H (Taster)      | (T) BC34 (m)   |
| I (Taster)           | auf LIN        |
| J (Taster)           | (R) SW         |
| K (Taster)           | (intern) SW    |
| L (Taster)           | (R) BC7        |
| M (Taster)           | (intern) SW    |
| N (Taster)           | (T) BC46 (40s) |
| (F3) O (Taster)      | auf LIN        |
| (F4) P (Taster)      | auf LIN        |
| Hupe                 | auf LIN        |



| Ref                                                                  | Description                        | Type* | Indicator |
|----------------------------------------------------------------------|------------------------------------|-------|-----------|
| A                                                                    | PTT 2                              | T     | -         |
| B                                                                    | Auxiliary Headlight                | T     | -         |
| C                                                                    | Alley Light – Left                 | R     | Blue      |
| D                                                                    | Alley Light – Right                | R     | Blue      |
| E                                                                    | Take Down Light – Steady Burn      | R     | Green     |
| F                                                                    | Take Down Light – Wig Wag          | R     | Red       |
| G                                                                    | Push-To-Talk                       | T     | -         |
| H                                                                    | Push-To-PA                         | T     | -         |
| I                                                                    | Cruise Light (US not X114A) no BCA | T     | Yellow    |
| J                                                                    | Headlight “OFF”                    | T     | Green     |
| K                                                                    | BC Speedometer Speed Lock          | T     | -         |
| L                                                                    | RH Radio Speaker Mute              | T     | Blue      |
| M                                                                    | Rear Two-Way Flashers              | T     | Red       |
| N                                                                    | Gun Lock “OPEN”                    | T     | -         |
| O                                                                    | Siren Activation                   | T     | -         |
| P                                                                    | Emergency Light Activation         | T     | -         |
| * (T)entative Switch (push-button) (R)ocker switch, stay in position |                                    |       |           |